Solid Rocket propellant internal ballistics

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Nomenclature

ρ_p	Propellant density
σ	Standard deviation c^*
σ_a	Standard deviation a
σ_n	Standard deviation n
a	Vieille law first parameter
A_b	Burning area
A_t	Nozzle throat area
c^*	Characteristic velocity
d_c	Chamber diameter
d_t	Nozzle throat diameter
L	Grain length
n	Vieille law second parameter
N_{it}	Number of iterations
P_c	Chamber Pressure
r_i	Grain internal radius

Acronyms

BATES Ballistic Test and Evaluation System. 1

BC Bayern Chemie. 1

HTPB Hydroxyl-terminated polybutadiene. 1

SRM Solid Rocket Motor. 1

Contents

1 Introduction

In this report, experimental data of a Ballistic Test and Evaluation System (BATES) motor [Figure 1] is analysed to derive Vieille's law and then a Monte Carlo analysis is performed and its results showcased. Firstly, the procedure through which the burning rate is obtained is presented, starting from experimental data related to the pressure curves of a BATES motor. The Bayern Chemie (BC) method is used to identify the effective pressure and burning rate. Pressure traces obtained from nine propellant batches tested at three pressure levels are analysed, yielding a total of 27 pressure traces. The pressure ranges related to the throat diameters shown in Table 1, are identified as follows: low pressure with a value around $30\,\mathrm{bar}$, a mid pressure around $40\,\mathrm{bar}$ and a high pressure around $50\,\mathrm{bar}$. The uncertainty in the model arises from the random variability in the production of these nine propellant batches, as well as firing-related uncertainties. Since the uncertainty is of aleatory nature, it can be represented by a Gaussian distribution, with an average value and a standard deviation. Subsequently, the coefficients of Vieille's law and the related uncertainties, given at σ, are derived. Finally, a Monte Carlo analysis is performed, which allows us to introduce the aleatory uncertainty coming from the experiments, expressed as uncertainty on the Vieille law coefficients and c^*, into the non-linear problem regarding the computation of the burning rate.

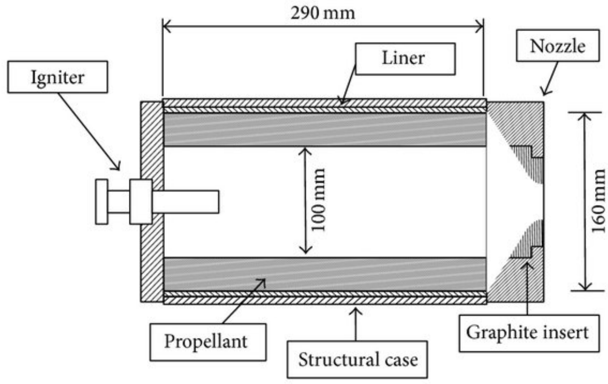

Figure 1: BARIA engine schematic.

	Low $\mathbf{P_c}$	**Mid $\mathbf{P_c}$**	**High $\mathbf{P_c}$**
Throat diameter [mm]	21.81	25.26	28.80

Table 1: BARIA engine characteristics.

The propellant batches are nominally identical to each other and composed as follows: 68% Ammonium Perchlorate (NH_4ClO_4), 18% Aluminium (Al) and 14% HTPB binder ($C_{7,075}H_{10,65}O_{0,223}N_{0,063}$). The density of the propellant is computed giving a value of $1762\,\mathrm{kg\,m^{-3}}$.

2 Ballistic analysis

In this Section, a ballistic analysis of the experimental data is performed. The objective is to characterize the whole propellant production through its Vieille's Law. To achieve this goal, the BC method is applied to compute the effective pressure trace [1]. The action of a Solid Rocket Motor (SRM) is effective when pressure traces cross the 5% of the maximum pressure, so that ignition and extinguishment transients are discarded. By identifying this reference pressure, the burning time and effective pressure can be retrieved following the method. Equivalently, the burning rate can also be computed from the ratio between web thickness and burning time.

Thanks to BC method, the effective pressure and burning rates of the 27 tests are obtained. From this information, the coefficients of the Vieille's Law, along with their standard deviations, can be computed through data regression. Table 2 shows the results for the coefficients, and Figure 2 compares the Vieille's Law fit, along with the experimental data points.

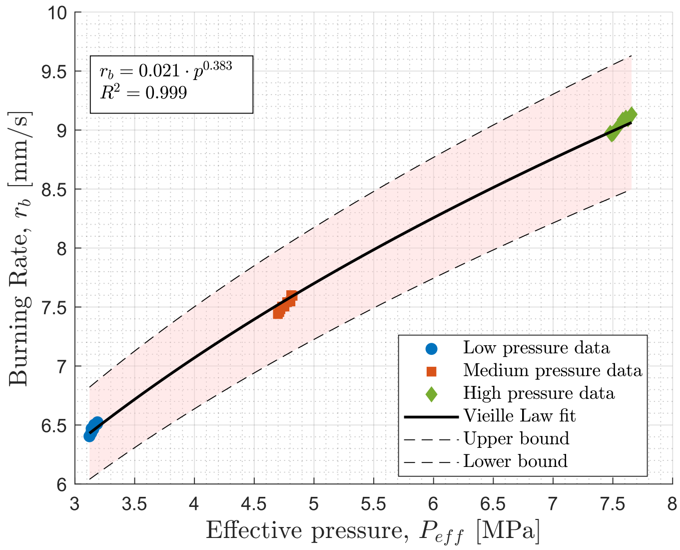

Figure 2: Burning rate vs effective pressure.

$\mathbf{a}$ $\left[\frac{mm}{s\ Pa^n}\right]$	$\sigma_{\mathbf{a}}$ $\left[\frac{mm}{s\ Pa^n}\right]$	% Deviation	$\mathbf{n}$ $[-]$	$\sigma_{\mathbf{n}}$ $[-]$	% Deviation
0.021041	0.000917	4.36	0.382659	0.002828	7.39

Table 2: Vieille's Law coefficients.

For each pressure curve, the experimental value of the characteristic velocity, c^*, is computed by integrating over time using a trapezoidal numerical integration scheme. Under the assumption that c^* is changing negligibly with respect to pressure, the entire data set is averaged to obtain a mean and an uncertainty, leading to results in Table 3.

c^* $[\mathrm{m\,s^{-1}}]$	σ_{c^*} $[\mathrm{m\,s^{-1}}]$	% Deviation
1512.61	12.16	0.80

Table 3: c^* mean and standard deviation.

3 Motor model

In this Section, a model is produced for the computation of a generic rocket firing. The motor will operate at low chamber pressure values, with the corresponding throat diameter of $d_t = 21.81\,\mathrm{mm}$. This model takes as inputs the Vieille's Law coefficients and characteristic velocity, returning burning time as an output. The model builds up a new pressure trace, which is dependent on the burning area along time:

$$p(t) = \left(a \rho_p c^* \frac{A_b(t)}{A_t} \right)^{\frac{1}{1-n}} \tag{3.1}$$

The burning area is computed accounting for a consumption of the grain along radial direction and along the grain axis from nozzle to igniter, resulting in:

$$A_b(t) = 2\pi r_i(t) L(t) + \frac{\pi}{4}\left(d_c^2 - (2r_i(t))^2\right) \tag{3.2}$$

Then, the burning time is retrieved as the total time elapsed between the ignition of the motor and the instant when the internal radius of the grain reaches the chamber wall. The burning time computed is $t_b = 4.62\,\mathrm{s}$.

4 Monte Carlo method

The ballistic properties of the propellant are used to carry out a Monte Carlo simulation to characterize the burning time. Randomized values for the parameters a and n of the Vieille's Law and the characteristic velocity c^* are generated according to their mean value and standard deviation and stored in a $N_{it} \times 3$ matrix. N_{it} is the number of iterations, considered to be the same for all three parameters. The order of the matrix rows, formed by parameter triplets, is randomly shuffled to get a proper convergence behaviour. For each triplet of values the burning time is computed as the time needed for the internal radius of the propellant to become equal to the chamber radius. This computation is achieved through a *while* cycle where at each time step, pressure, burning rate, internal radius, propellant length and burning area are updated.

5 Results and conclusions

The results of the Monte Carlo analysis are displayed in Figure 3 and Figure 4, showing the typical plateau behaviour of this kind of simulation.

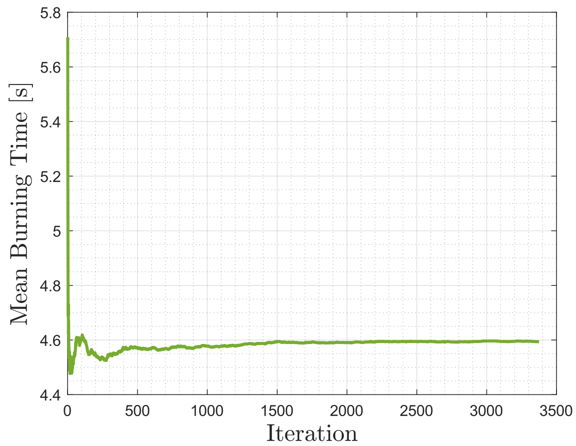

Figure 3: Mean burning time.

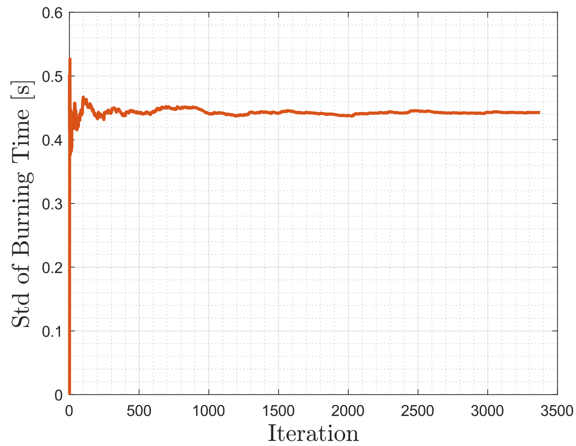

Figure 4: Burning time standard deviation.

From Figure 3, it is evident that the curve quickly reaches a plateau after only a few iterations, confirming the stability and convergence of the model. Regarding Figure 4, the value reached by the standard deviation is rather high. It is therefore important to understand which of the three uncertain input variables has more impact on the final uncertainty of the burning time. Due to the smaller relative standard deviation of the c^* with respect to the other variables, it can be inferred that it has an almost negligible effect. This hypothesis is confirmed after running a Monte Carlo analysis with fixed c^*.

References

[1] *Flipped class on SRM internal ballistics: The Bayern Chemie method (BC)*. Lecture notes, Politecnico di Milano. 2025.

[2] George P. Sutton and Oscar Biblarz. *Rocket Propulsion Elements*. 9th ed. John Wiley & Sons, 2016. ISBN: 978-1-118-75365-1.